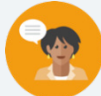


13.1 Evaluating Expressions

Lesson Introduction

 Benchmark	MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations.		 Learning Targets	- I can apply order of operations to an algebraic expression. - I can evaluate expressions that include exponents.
 Benchmark Coherence	Building On MA.5.AR.2.1 Translate written real-world and mathematical descriptions into numerical expressions and numerical expressions into written mathematical descriptions.		Working Toward MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.	
 Essential Questions	- How do you evaluate an expression by applying the order of operations? - How do you evaluate an expression when it includes an exponent? - What information is needed in order to “evaluate” an algebraic expression?			
 Lesson Narrative	In this first lesson of the unit, students extend their understanding of evaluating numeric expressions using the order of operations to evaluate algebraic expressions. Students connect numeric expressions with and without grouping symbols to mathematical models, deepening their understanding of these symbols. Students were introduced to evaluating exponents in Unit 6 and evaluating numeric expressions involving exponents in Unit 11. This lesson will be the first time they are evaluating exponents within algebraic expressions that include multiple operations.			
 Component Summary	13.1.1 Warm-Up (~5 min.)	Evaluate a numeric expression to resolve a disagreement (<i>SE p. 263</i>)	13.1.4 Guided Instruction (10 min.)	Evaluate numeric expressions with exponents (<i>SE p. 267</i>)
	13.1.2 Guided Instruction (15 min.)	Extend evaluating numeric expressions using the order of operations to algebraic expressions (<i>SE p. 264</i>)	13.1.5 Try It (10 min.)	Evaluate algebraic expressions with exponents (<i>SE p. 268</i>)
	13.1.3 Collaboration (5-10 min.)	Explore how exponents can be written out as products (<i>SE p. 266</i>)	13.1.6 Wrap-Up (~5 min.)	Find and explain errors in students’ work (<i>SE p. 269</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Order of operations - Variable - Evaluate - Value - Parentheses Additional Terms: N/A		(optional) Colored pencils	
			Additional Preparation	
			N/A	

	Common Misconceptions	Things to Consider	
 Teacher Tips	- Students may substitute the variable as a digit instead of multiplying it by the coefficient. For example, in $5x$, where $x = 4$, students may write 54 instead of $5(4)$. - Students may prioritize left to right operations in lieu of parenthesis. For example, during the second expression in 11.1.2 Collaboration, $3 \times (5 + 4)$, students may multiply 3×5 first.	- This lesson has several key glossary terms necessary for comprehension. Consider having students add to interactive notebooks or create a Frayer Model for key vocabulary. - Consider modeling examples and allowing students to demonstrate their mastery with chips to make sense of expressions. - There are many misconceptions about the order of operations. The teaching of the mnemonic device of PEMDAS leads students to perceive that the order of PEMDAS is rigid, where it is not. For example, in the expression $30 - 4(3 + 7) + 12 \div 3$, there are options as to where to begin. Students actually have a choice and may first simplify the $3 + 7$ in the parentheses, distribute the 4 to the 3 and to the 7, or perform $12 \div 3$ before doing any other computation—all without affecting an accurate outcome. For this reason, the lessons do not empathize this mnemonic device. Encourage your students to consider different entry points and discuss them. - If time is an issue, consider assigning part of all of the 13.1.5 Try It for homework.	
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard	
	Compare and Connect For 13.1.5 Try It, consider using the Compare and Connect routine by encouraging students to compare how the values of the expressions change depending on the order in which the operations are performed.	MA.K12.MTR.4.1: Discussion and Reflection 13.1.2 Guided Instruction opens by presenting students with three hypothetical evaluations of an expression and calls on them to find which evaluation is correct. Clarifications for this standard include recognizing an error and suggesting how to fix it. Encourage students to not only find the error but formalize how to prevent it from happening	
 Differentiate Instruction	For 13.1.5 Try It, consider placing students in differentiated groups during this activity. Students who have demonstrated a good grasp of the content can be assigned to work independently to complete the questions, while students who may still be struggling can work through the content with guidance from the teacher before trying independently.		
Lesson Notes:			

13.1.1 Warm-Up: Evaluating Numeric Expressions

Component Overview: Students examine two different values to determine which one correctly represents the expression, then work to find the value of another expression independently.



13.1.1 Warm-Up: Evaluating Numeric Expressions

This math expression is dividing the internet.

What is the value of $8 \div 2(2 + 2)$?



Poll the Class

Engage students with a Poll the Class routine to determine whose answer they agreed with. Consider having students write their responses on a whiteboard and holding it up or show either one or two (using their hand) to indicate who they agree with. Encourage students to discuss how they know who is correct and the error the other person made.

- Do you agree with Alex or Ms. B? Explain or show why using order of operations.

Ms. B

$$8 \div 2(2 + 2)$$

$$8 \div 2(4)$$

$$4(4)$$

$$16$$

- Find the value of the expression. Show your work.

$$4 - 2 \times 6 \div 3 + 7$$

$$4 + (-2) \times 6 \div 3 + 7$$

$$4 + (-12) \div 3 + 7$$

$$4 + (-4) + 7$$

$$4 - 4 + 7$$

$$0 + 7$$

$$7$$



Students may not have social media, but they could recreate the activity for homework by polling their friends and family.

13.1.2 Guided Instruction: Applying Order of Operations to Algebraic Expressions

Component Overview: Activate students' prior knowledge both from previous units and from this lesson in the 13.1.1 Warm-Up. Position students as sense makers in this error analysis scenario in order to elicit increasingly precise and mathematical responses on evaluating expressions.



13.1.2 Guided Instruction: Applying Order of Operations to Algebraic Expressions

Recall from 5th grade that an order of operations is used to evaluate numeric expressions with multiple operations. Grouping symbols such as parentheses are used to indicate which operations should be performed first. The following order is used.

- Step 1: Perform operations within grouping symbols.
- Step 2: Evaluate exponents.
- Step 3: Multiply and divide in order from left to right ($\rightarrow$).
- Step 4: Add and subtract in order from left to right ($\rightarrow$).

1. Three students completed the warm-up. The steps in each student's process are shown.

Leila's Work	Fabian's Work	Jaylah's Work
$4 - 2 \times 6 \div 3 + 7$	$4 - 2 \times 6 \div 3 + 7$	$4 - 2 \times 6 \div 3 + 7$
$2 \times 6 \div 3 + 7$	$4 - 12 \div 3 + 7$	$4 - 12 \div 3 + 7$
$12 \div 3 + 7$	$4 - 4 + 7$	$4 - 4 + 7$
$4 + 7$	$4 - 11$	$0 + 7$
11	-7	7

- a. Complete the statements about which student correctly applied the given steps of the order of operations.

☐ Leila
☐ Fabian
☒ Jaylah

evaluated the numerical expression correctly.

She first performed all ☒ multiplication and division ☐ addition and subtraction in

order from left to right and then performed all

☐ multiplication and division
☒ addition and subtraction

in order from left to right.



Error Analysis

Engage students in an error analysis of each student's work from question 1. Consider having students partner up to analyze the student work and determine the errors before participating in a whole class discussion to address the misconceptions displayed in each response.



Challenge students to not only name the errors each student made but to also write an explanation for how to fix and prevent them from making the same mistakes in the future.

13.1.2 Guided Instruction: Applying Order of Operations to Algebraic Expressions (continued)

- b. With your partner, discuss the work of each student, Leila, Fabian, and Jaylah. Two of these students made a mistake somewhere. Circle the mistakes that you and your partner found.

Leila went from left to right.

Fabian did addition before subtraction.

2. Omari evaluated Expression A and Expression B. His work is shown.

Expression A	Expression B
$12 + 24 \div 6 - 2 \times 2$	$12 + 24 \div (6 - 2 \times 2)$
$12 + 4 - 2 \times 2$	$12 + 24 \div (6 - 4)$
$12 + 4 - 4$	$12 + 24 \div (2)$
$16 - 4$	$12 + 12$
12	24

- a. With your partner, discuss the two original expressions that Omari evaluated and his corresponding answers.
- b. Omari's partner gave him a thumbs up on both of his answers. Why did Omari arrive at two different values when evaluating these expressions?
 Omari had two different answers because Expression B contains parentheses that are not in Expression A. Following order of operations Expression B is completed differently than Expression A.

Like numeric expressions, algebraic expressions can be evaluated using order of operations when given a specific value for the **variable(s)**.

In an algebraic expression, a **variable** is a letter that represents a number. It is called a **variable** because the value used to replace the letter can **vary**.



Turn and Talk

Before answering question 2, have students participate in a Turn and Talk routine to formalize their answers before circling the mistakes. Encourage students to notice the differences between the two expressions and consider how/why that might impact the value of the expressions.



- What is the same about the expressions? What is different?
- How do parentheses impact the order of operations? How do they impact the value of the expression?



Consider having students create a foldable, Frayer Model, or other type of interactive to allow students to synthesize the vocabulary outlined in this section. Examples, nonexamples, and verbal descriptions will provide multiple pathways for students to engage with the content in meaningful ways that encourage conceptual understanding over rote memorization.

13.1.2 Guided Instruction: Applying Order of Operations to Algebraic Expressions (continued)

3. When evaluating algebraic expressions, first substitute the given **number** for the variable(s). Then, apply the given steps for order of operations.



In mathematics, **evaluate** is an instruction that indicates the answer is a **value**.

4. Evaluate $\frac{n}{6} + 2$ when $n = 12$.

$$\frac{12}{6} + 2 = 2 + 2 = 4$$

In numeric expressions, multiplication can be represented in different ways. Each expression below is a way to represent "three times five."

$$3 \times 5$$

$$3 \cdot 5$$

$$3(5)$$

In algebraic expressions, the multiplication symbol, $\times$, could be confused with the variable x . Therefore, writing a number next to a variable is a better way to note multiplication when using variables.

$3x$ means "three times x ."

Multiplication of a number and a variable could also be represented using the multiplication dot, $3 \cdot x$, or parentheses, $3(x)$, but $3x$ will be the most common way.

5. Explain why $3 \cdot 5$ cannot be written as 35.

The dot in the problem represents multiplication. You need to multiply 3 times 5.

6. What is the value of $3x$ if $x = 5$?

$$3 \cdot 5 = 15$$



When substituting values into algebraic expressions, it is helpful to put the substituted value in **parentheses** when the variable is being multiplied by a known value.

7. Find the value of $7m - (3 + b - 12) \div 2$ if $m = 4$ and $b = 15$. Show your work.

$$7(4) - (3 + 15 - 12) \div 2$$

$$28 - (18 - 12) \div 2$$

$$28 - (6) \div 2$$

$$28 - 3 = 25$$



- How do the parentheses impact the evaluation of the expression?
- How would the value change if the parentheses were moved around the first part of the expression?

13.1.3 Collaboration: Order of Operations

Component Overview: Read the red boxes out loud to students, then have them work in partner pairs to find the numeric value of the given expression. After reviewing their work, engage the class in a discussion about the differences between their work and what is presented.



13.1.3 Collaboration: Order of Operations



An order of operations was first noted in First Year Algebra by Webster Wells and Walter Wilson Hart. The textbook was published in 1912.

- The problem below is an example from First Year Algebra.

EXAMPLE. Find the numerical value of the expression,

$$4ab + \frac{6c}{b} - d^3,$$

when $a = 4$, $b = 3$, $c = 5$, $d = 2$.

- What is the numerical value of the expression? Show your work.

$$\begin{aligned} &4(4)(3) + \frac{6(5)}{3} - 2^3 \\ &48 + \frac{30}{3} - 8 \\ &48 + 10 - 8 \\ &50 \end{aligned}$$

- Check your work with your partner.



Notice and Wonder

Consider implementing a Notice and Wonder routine for question 1. Ask students to identify what they notice and wonder about their work and the work from the textbook.



What are some differences between numeric and algebraic expressions? The textbook changed d^3 to $2 \cdot 2 \cdot 2$. The order of operations in the textbook did not include exponents, so each exponent was rewritten as multiplication.

- When evaluating numeric expressions, how many values are possible?
- When evaluating algebraic expressions, how many values are possible?
- What information is needed in order to evaluate an algebraic expression?

13.1.3 Collaboration: Order of Operations

c. The work shown in the textbook is below.

$$4ab + \frac{6c}{b} - d^3 = 4 \cdot 4 \cdot 3 + \frac{6 \cdot 5}{3} - 2^3 = 4 \cdot 4 \cdot 3 + \frac{6 \cdot 5}{3} - 2 \cdot 2 \cdot 2$$

$$= 48 + 10 - 8 = 50.$$

What is the difference between your work and the work from the book?

Emphasis should be placed on writing 2^3 as $2 \cdot 2 \cdot 2$. This is an illustration of how the order of operations was originally written without exponents. Instead, it was written out as a product.



An order of operations became more important when programmers needed a set of rules to program a computer to find values of numeric expressions. In 1883, Ada Lovelace described in her book about Charles Babbage's analytical engine, the first computer, that codes could be created so the device could handle letters and symbols along with numbers.



Convey to students that the original computers were what is known today as calculators. Their only purpose was to do mathematical operations. Students who have seen *Hidden Figures* may be able to relate to this because the Katherine Johnson, Mary Jackson, and Dorothy Vaughn were referred to as computers. In the film, their jobs appeared to be threatened by the installment of computers which were machines that would do the calculations they were doing by hand.

13.1.4 Guided Instruction: Evaluating Expressions with Exponents

Component Overview: Guide students through the instructions at the beginning of this component. Engage students in a discussion about why evaluating exponents should be done before other multiplication and division in the expression. Then have students work independently to evaluate the numeric expressions. Finally, engage students in a discussion to debrief the questions and consolidate their understandings.



13.1.4 Guided Instruction: Evaluating Expressions with Exponents

The work in the book, First Year Algebra, showed 2^3 written out as multiplication. This method can get complex as shown in the table.

$(-3)^3 + 2^6 \div -16 + 5$
Exponents written as multiplication
$((-3)(-3)(-3)) + (2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2) \div -16 + 5$

$(-3)^3 + 2^6 \div -16 + 5$
Evaluate expressions with exponents first
$-27 + 64 \div -16 + 5$

- When writing $(-3)^3$ and 2^6 using multiplication, both were written inside a set of parentheses.
- In order of operations, parentheses indicate that the operations inside should be done **before** other steps.
- Therefore, evaluating exponents is done before other multiplication and division when following the order of operations.

- Evaluate $(-3)^3 + 2^6 \div -16 + 5$.

$$\begin{aligned} & -27 + 64 \div -16 + 5 \\ & -27 + (-4) + 5 \\ & -31 + 5 = -26 \end{aligned}$$

- Evaluate each expression, then determine if the expressions are equivalent.

Expression A	Expression B	Equivalent?
$(-3)^4 = 81$	$-(3^4) = -81$	No
$-(9)^2 = -81$	$(-9)^2 = 81$	No
$(-5^2) = -25$	$(-5)^2 = 25$	No
$26 - 4^3$ $= 26 - 64$	$26 - (-4)^3$ $= 26 + 64$	No
$17 + (-2^4)$ $= 17 + (-16)$	$17 - (2^4)$ $= 17 - 16$	Yes



This activity provides an entry point for visual learners through the use of color-coding the expressions in the table.



Students may need a brief review of evaluating exponents, particularly with negative bases. The second example in 13.1.4 Guided Instruction provides an opportunity to monitor to assess formatively how the students approach this within the order of operations. Guide students in order to address common misconceptions as needed with specific students or the whole class through modeling.

13.1.4 Guided Instruction: Evaluating Expressions with Exponents (continued)

3. Describe a strategy that can be used when evaluating expressions with exponents that include a negative sign or a subtraction sign.

Answers will vary. Guide students to even/odd exponents, the placement of a negative in front of the parentheses as opposed to inside.

4. Elizabeth evaluated $b + a^3 \cdot 7 - 9$ when $a = -4$ and $b = 17$. Her work is below.

- a. Circle Elizabeth's error.

$$17 + (-4)^3 \cdot 7 - 9$$

$$17 + (-64) \cdot 7 - 9$$

$$(-47) \cdot 7 - 9$$

$$-329 - 9$$

$$-338$$

- b. Which step of the order of operations did she not apply correctly?
- Ⓐ Evaluate expressions with exponents.
 - Ⓑ Perform operations inside grouping symbols.
 - Ⓒ Perform multiplication and division in order from left to right.
 - Ⓓ Perform addition and subtraction in order from left to right.



The order of performing computations is first to work within grouping symbols using the order of operations. Then, simplify terms with exponents. Next, while reading from left to right, perform multiplication and division in the order in which it appears. Finally, while reading from left to right, perform addition and subtraction in the order in which it appears.



Solidify understanding by asking these questions:

- What are some differences between numeric and algebraic expressions? (Numeric expressions contain all known values while algebraic expressions include some unknown value(s).)
- When evaluating numeric expressions, how many answers are possible?
- What information is needed in order to "evaluate" an algebraic expression? (Specific values for the variable(s).)



While reviewing the order of operations in the vocabulary box, remind students that up to this point it has been shown that these are not hard and fast rules but should be used to guide them when they determine a value. Although there are some things where the order does not matter there are others where the order does matter. It may be helpful to create an anchor chart or reference cards with examples for students to refer to as they build their understanding.

13.1.5 Try It: Discovering the Importance of Left to Right

Component Overview: Students work independently to evaluate expressions in different ways, working to determine how changing grouping symbols or the order in which operations are performed changes the value of the expression. After discussing question 1, use your discretion to determine if students can work independently or would do better checking in with a partner regularly.



13.1.5 Try It: Discovering the Importance of Left to Right

1. Mia is evaluating the expression, $x^3 \div r \cdot (k^2 - 5)$ for $k = 3, r = -8$ and $x = 4$. Her first few steps are shown:

$$\begin{aligned} 4^3 \div -8 \cdot (3^2 - 5) \\ 64 \div -8 \cdot (9 - 5) \\ 64 \div -8 \cdot 4 \end{aligned}$$

- a. Mia thinks that she could finish her work by either finding $64 \div -8$ or $-8 \cdot 4$ first. Show the work for both possibilities to determine if Mia is correct.

$$\begin{array}{ll} 64 \div -8 \cdot 4 & 64 \div -8 \cdot 4 \\ (64 \div -8) \cdot 4 & 64 \div (-8 \cdot 4) \\ -8 \cdot 4 = -32 & 64 \div -32 \\ & -2 \end{array}$$

The answers are not the same.

- b. Was Mia correct when she stated that both methods would work? Use the steps of the order of operations to justify whether or not Mia is correct.

Multiplication and division should be done from left to right.

2. Answer the following problems.

- a. Evaluate $r \cdot W \div x \div m$ when $m = 2, r = -4, W = -60$ and $x = -10$. Show your work.

$$\begin{aligned} -4 \cdot (-60) \div -10 \div 2 \\ 240 \div -10 \div 2 \\ -24 \div 2 \\ -12 \end{aligned}$$



Compare and Connect

Each question in this section provides an opportunity for students to compare how changing the order in which operations are performed changes the value of the expression. Consider modeling a Think Aloud for question 1b to model the thinking students should be doing. Then have students engage in this Compare and Connect with a partner before completing question c for 2.



Consider splitting the class into differentiated groups to complete this section after modeling the Think Aloud to answer question 1. Guide students who may need more support through the thinking for question c.

13.1.5 Try It: Discovering the Importance of Left to Right (continued)

- b. Evaluate $(r \cdot W) \div (x \div m)$ when $m = 2$, $r = -4$, $W = -60$ and $x = -10$. Show your work.

$$(-4 \cdot -60) \div (-10 \div 2)$$

$$240 \div -5$$

$$-48$$

- c. How did the use of parentheses change the answers?

The parentheses changed the order so that $-10 \div 2$ was divided first instead of $-240 \div -10$ being first.

The parentheses around $r \cdot W$ did not change what was done without parentheses.



- What happens when you use parentheses in the expression? How do they change the value of the expression? Why does it change the value?
- How does the value of the expression change when you rearrange the order of the operations?

13.1.6 Wrap-Up: Error Analysis

Component Overview: In this formative assessment, students demonstrate their understanding of evaluating algebraic expressions. Students analyze two students' work to identify errors. Then they explain how to fix the errors in a note to the students.



13.1.6 Wrap-Up: Error Analysis

- Nadine and Nora evaluated the expression $r^3 \div x \cdot c$ for $c = 9$, $r = -6$ and $x = -3$. Both students made errors in their work shown. Circle their errors.

Nadine's Work	Nora's Work
$r^3 \div x \cdot c$ $(-6)^3 \div (-3) \cdot 9$ $-216 \div (-3) \cdot 9$ $-216 \div -27$ 8	$r^3 \div x \cdot c$ $(-6)^3 \div (-3) \cdot 9$ $-18 \div (-3) \cdot 9$ $6 \cdot 9$ 54



Consider allowing students to record their answers using an app like FlipGrid. This will be beneficial for students who may understand the content but struggle with written response.

- Write a short explanation to each student to explain the mistake and what she needs to remember about evaluating expressions in the future.

To Nadine: Nadine, The order of operations states to multiply and divide from left to right. In the step $-216 \div (-3) \cdot 9$ you should divide $-216 \div (-3)$ then multiply $72 \cdot 9$.	To Nora: Nora, $(-6)^3$ is a shortcut way of writing $-6 \cdot -6 \cdot -6$, where: $-6 \cdot -6 = 36$ $36 \cdot -6 = -216$
--	--